

Internship Report

On

Machine Learning Internship at CodeAlpha:

**Handwritten Character Recognition using CNN and Speech Emotion
Recognition using Deep Learning**

Submitted To

School of Computer Science and Engineering

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In partial fulfilment of the requirement of the degree of
B.Tech CSE

By

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Batch: 2026-2027

2026-27

INTERNSHIP COMPLETION CERTIFICATE

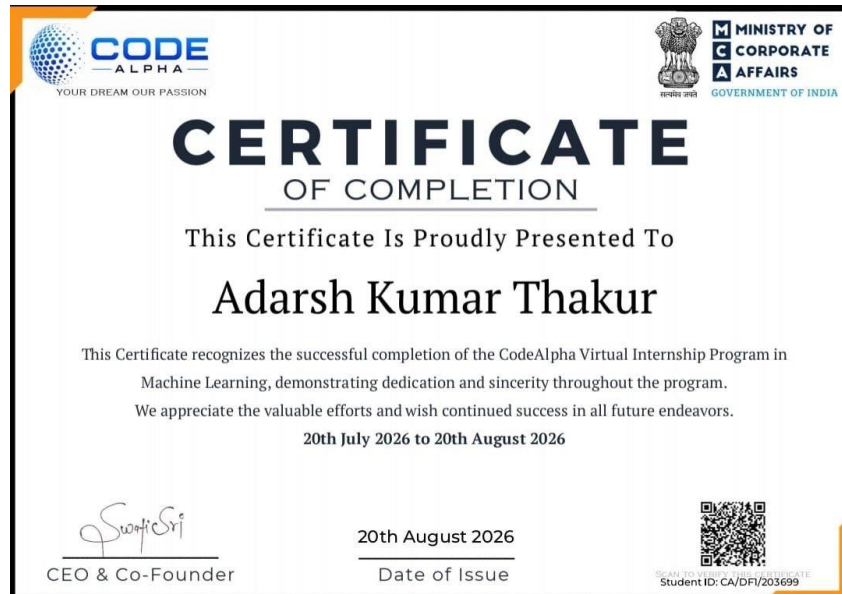


Figure 3.1: CodeAlpha Internship Completion Certificate awarded to Adarsh Kumar Thakur

CANDIDATE'S DECLARATION

I, Adarsh Kumar Thakur, do hereby declare that the internship report titled “Machine Learning Internship at CodeAlpha: Handwritten Character Recognition using CNN and Speech Emotion Recognition using Deep Learning” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references used have been duly quoted, and I have not taken as such any material from any other source without acknowledgement.

I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: Adarsh Kumar Thakur

Roll No.: CS-2341736

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Machine Learning Internship Program at CodeAlpha. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during the course of this work.

I am sincerely grateful to the mentors and team at CodeAlpha for sharing their truthful and illuminating views on a number of issues related to Machine Learning and Deep Learning, and for giving me the opportunity to work on practical, industry-relevant projects that strengthened my technical and analytical skills.

I would also like to thank the School of Computer Science and Engineering, IILM University, Greater Noida, for providing me the platform and encouragement to undertake this internship as part of my B.Tech curriculum.

Finally, I express my heartfelt gratitude to my parents and family for their constant support, encouragement, and blessings throughout the duration of this internship.

Signature: _____

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Roll No.: CS-2341736

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a summary of the work undertaken during my one-month Virtual Internship Program in Machine Learning at CodeAlpha, carried out from 20th July 2026 to 20th August 2026. The internship provided hands-on, project-based exposure to core concepts of Machine Learning (ML) and Deep Learning (DL), including data preprocessing, model building, evaluation, and deployment of trained models into interactive web applications.

During the internship, I successfully completed two independent projects: (i) Handwritten Character (Digit) Recognition using a Convolutional Neural Network (CNN) trained on the MNIST dataset, and (ii) Speech Emotion Recognition (SER) using Deep Learning, which analyses speech audio and classifies the emotional characteristics expressed in the voice. Both projects involved the complete machine learning pipeline — from data acquisition and preprocessing, through model design, training and evaluation, to deployment as interactive web applications using Streamlit and Flask respectively.

This internship enabled me to apply theoretical concepts learned during my B.Tech coursework to real, practical problems, and helped me strengthen my skills in Python programming, deep learning frameworks such as TensorFlow and Keras, image and audio signal processing, model evaluation techniques, and the deployment of machine learning models as usable applications.

4.2 Organization Profile

CodeAlpha is a technology-focused organization that offers virtual internship programs to students and early-career professionals in domains such as Machine Learning, Web Development, Android Development, Data Science, Python Programming, and Cyber Security. The organization's mission, reflected in its tagline “Your Dream, Our Passion”, is to bridge the gap between academic learning and industry requirements by offering practical, project-based internship experiences that help learners build real-world skills and a stronger professional portfolio.

As part of its Machine Learning Internship track, CodeAlpha provides a list of project tasks from which interns select and independently implement a defined number of projects within a specified duration. On successful completion, CodeAlpha issues an official Certificate of Completion (recognized with reference to the Ministry of Corporate Affairs, Government of India) along with a Letter of Recommendation, enabling interns to showcase their practical skills and internship experience to recruiters and academic institutions.

4.3 Problem Statement

Project 1 – Handwritten Character Recognition: Recognizing handwritten digits automatically is a classical yet practically important problem in the field of computer vision, with applications in postal mail sorting, bank cheque processing, digitized form reading, and optical character recognition (OCR) systems. Handwriting varies significantly from person to person in terms of shape, size, slant, and stroke thickness, which makes rule-based recognition unreliable. The problem addressed in this project was to design and train a deep learning model that can accurately classify a handwritten digit image (0–9) into its correct numerical class, despite this natural variability in handwriting.

Project 2 – Speech Emotion Recognition: Human speech carries not only linguistic content but also emotional information conveyed through tone, pitch, intensity, and rhythm. Automatically detecting emotion from speech has significant applications in customer-service call analysis, virtual assistants, mental-health monitoring tools, and human-computer interaction systems. The problem addressed in this project was to build a deep-learning-based system capable of analyzing raw speech audio and correctly classifying the emotional state expressed in it, despite challenges such as variation in speakers' voices, differing recording conditions, and background noise.

4.4 Project Objectives

The key objectives defined for the two internship projects were as follows:

Project 1: Handwritten Character Recognition using CNN

- To design and implement a Convolutional Neural Network (CNN) capable of recognizing handwritten digits from image data.

- To preprocess raw handwritten digit images using OpenCV so that they are suitable as input to the CNN model.
- To train and validate the model on the MNIST dataset and achieve a high classification accuracy on unseen test data.
- To evaluate the trained model using standard classification metrics such as precision, recall, F1-score, and a confusion matrix.
- To deploy the trained model as an interactive web application using Streamlit, allowing a user to upload an image and receive a predicted digit along with a confidence score.

Project 2: Speech Emotion Recognition using Deep Learning

- To develop a deep-learning-based system capable of analyzing speech audio and classifying the emotion expressed in it.
- To extract meaningful audio features (such as MFCCs – Mel-Frequency Cepstral Coefficients) that represent the emotional characteristics of speech.
- To design, train, and evaluate a deep learning classification model for emotion recognition from extracted audio features.
- To integrate the trained model with a Flask-based web application so that a user can upload a speech/audio file and view the predicted emotion along with its probability breakdown.
- To understand and document the real-world challenges involved in speech emotion recognition, such as variation in voice, recording conditions, background noise, and the inherent limitations of classifying emotion from speech alone.

4.5 Scope of the Project

The scope of the Handwritten Character Recognition project is limited to the classification of single, isolated handwritten digits (0–9) provided as an input image, using a model trained on the MNIST benchmark dataset. It does not cover recognition of full handwritten sentences, cursive handwriting, or alphabetic characters, but the underlying CNN architecture and preprocessing pipeline can be extended in future work to support such multi-character or multi-class recognition tasks.

The scope of the Speech Emotion Recognition project is limited to classifying a fixed, pre-defined set of emotions from short speech/audio clips uploaded by the user, based on acoustic features extracted from the audio signal. The system's accuracy is dependent on the quality of the audio recording and the training data used; it is not designed to perform speaker identification, speech-to-text transcription, or multilingual emotion analysis in its current form, though the architecture is extensible for such enhancements.

Both projects were developed for educational and demonstrative purposes within the CodeAlpha internship framework, with a strong emphasis on understanding and implementing the complete machine learning development lifecycle rather than building production-scale systems.

4.6 Technologies and Tools Used

The table below summarizes the key technologies, libraries, and tools used across the two projects completed during the internship.

Category	Project 1: Handwritten Digit Recognition	Project 2: Speech Emotion Recognition
Programming Language	Python	Python
Deep Learning Framework	TensorFlow & Keras (CNN)	TensorFlow & Keras
Data / Feature Processing	OpenCV, NumPy	Librosa (MFCC extraction), NumPy, Pandas
Model Evaluation	Scikit-learn (precision, recall, F1-score, confusion matrix)	Scikit-learn, Matplotlib
Dataset	MNIST Handwritten Digit Dataset (10,000 test images)	Labelled speech/audio emotion dataset
Web Deployment Framework	Streamlit	Flask

Table 4.1: Technologies and Tools used in Project 1 and Project 2

4.7 System Architecture

Project 1: Handwritten Digit Recognition – Architecture

The system follows a standard image-classification pipeline. A handwritten digit image, uploaded by the user through the Streamlit interface, is first pre-processed using OpenCV (grayscale conversion, resizing to 28×28 pixels, and normalization) to match the input format expected by the MNIST-trained model. The pre-processed image is

then passed through a Convolutional Neural Network consisting of convolutional layers (for feature extraction), pooling layers (for dimensionality reduction), and fully connected dense layers (for classification), followed by a softmax output layer that produces a probability distribution over the ten digit classes (0–9). The class with the highest probability is returned as the predicted digit, along with its confidence score.

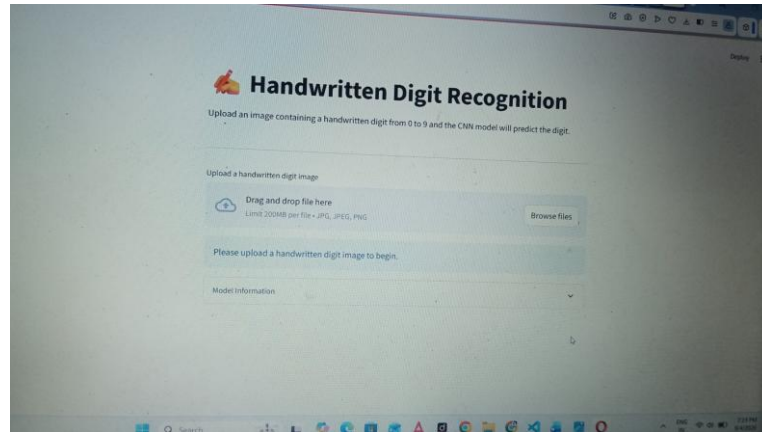


Figure 4.1: System Architecture / Working Interface – Handwritten Digit Recognition (Streamlit application, upload → preprocessing → CNN prediction)

Project 2: Speech Emotion Recognition – Architecture

The Speech Emotion Recognition system follows an audio-classification pipeline. A speech/audio file (.wav) uploaded by the user through the Flask web interface is first processed to extract acoustic features, primarily Mel-Frequency Cepstral Coefficients (MFCCs), using the Librosa library. These extracted features are normalized and passed as input to a trained deep learning classification model, which outputs a probability distribution over the possible emotion classes. The Flask backend (app.py) loads the trained model and associated artifacts through a dedicated predict module (predict.py, using `load_artifacts` and `predict_emotion` functions), performs inference on the uploaded file, and renders the predicted emotion along with a full probability breakdown back to the user through the web interface.

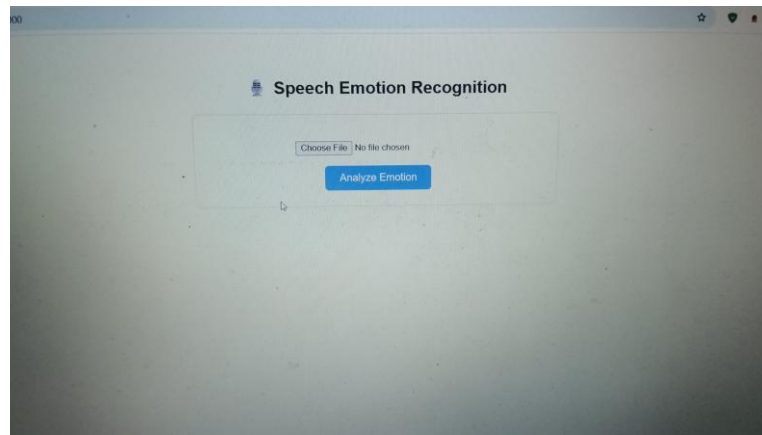


Figure 4.2: Working Interface – Speech Emotion Recognition (Flask application: file upload and “Analyze Emotion” action)

4.8 Methodology

Project 1: Handwritten Digit Recognition – Methodology

- Step 1: Data Collection: The standard MNIST dataset of handwritten digit images was used, consisting of labelled training images and 10,000 unseen test images.
- Step 2: Data Preprocessing: Images were converted to grayscale, resized, and normalized using OpenCV and NumPy to prepare them for input into the CNN.
- Step 3: Model Building: A Convolutional Neural Network was designed using TensorFlow and Keras, comprising convolutional, pooling, and dense layers suited to image classification.
- Step 4: Model Training: The CNN was trained on the training partition of the MNIST dataset, with the model's parameters optimized to minimize classification loss.
- Step 5: Model Evaluation: The trained model was evaluated on 10,000 unseen MNIST test images using accuracy, precision, recall, F1-score, and a confusion matrix (via Scikit-learn).
- Step 6: Deployment: The final trained model was integrated into an interactive Streamlit web application, allowing a user to upload a digit image and receive a real-time prediction with a confidence score.

Project 1: Model Performance

Metric	Value
Test Accuracy	99.11%

Metric	Value
Test Loss	0.0298
Macro F1-Score	0.9910
Test Dataset Size	10,000 unseen MNIST images

Table 4.2: Model Performance Metrics – Handwritten Digit Recognition

The trained CNN model achieved a test accuracy of 99.11% on 10,000 unseen MNIST test images, with a test loss of 0.0298 and a macro F1-score of 0.9910, demonstrating strong and consistent performance across all ten digit classes.

Project 2: Speech Emotion Recognition – Methodology

- Step 1: Data Collection: A labelled dataset of speech/audio samples, each associated with a specific emotional category, was used for training and evaluation.
- Step 2: Feature Extraction: Meaningful acoustic features, primarily MFCCs (Mel-Frequency Cepstral Coefficients), were extracted from each audio sample using the Librosa library to numerically represent the emotional characteristics of the speech signal.
- Step 3: Model Building: A deep learning classification model was designed using TensorFlow and Keras to map the extracted audio features to emotion categories.
- Step 4: Model Training: The model was trained on the extracted features, with model parameters optimized to correctly classify the emotion expressed in each audio sample.
- Step 5: Model Evaluation: The trained model's performance was evaluated using classification metrics available through Scikit-learn, and results were visualized using Matplotlib.
- Step 6: Deployment: The trained model and its preprocessing artifacts were integrated into a Flask web application (app.py and predict.py) that allows a user to upload a .wav audio file and view the predicted emotion along with a probability breakdown for each emotion class.

```

1  #!/usr/bin/env python
2  # coding: utf-8
3
4  """
5  Single Flask web demo: upload a .wav file in the browser and get back
6  the predicted emotion with a probability breakdown.
7  """
8
9  # Import Flask and other dependencies
10 from flask import Flask, request, jsonify, render_template
11
12 # Import the model
13 from predict import load_artifacts, predict_emotion
14
15 # Create the Flask application
16 app = Flask(__name__)
17
18 # Load the model and artifacts
19 model, encoder, norm_stats = load_artifacts()
20
21 # Define the main route
22 @app.route('/', methods=['GET', 'POST'])
23 def index():
24     if request.method == 'POST':
25         # Get the audio file from the request
26         file = request.files.get("audio_file")
27         if file:
28             # Predict the emotion
29             prediction = predict_emotion(file)
30             # Return the prediction as JSON
31             return jsonify(prediction)
32     else:
33         # Return the index template
34         return render_template("index.html")
35
36 # Run the application
37 if __name__ == '__main__':
38     app.run(debug=True)

```

Figure 4.5: Flask application source code (app.py) implementing the Speech Emotion Recognition web demo

4.9 Expected Outcomes

Both projects were completed successfully, achieving the following outcomes:

- A fully functional Handwritten Digit Recognition system that accurately classifies handwritten digit images with a test accuracy of 99.11% and provides prediction confidence scores through an interactive Streamlit web interface.
- A functional Speech Emotion Recognition system capable of analyzing uploaded speech audio and classifying the expressed emotion with associated confidence/probability scores, deployed through a Flask web application.
- A strengthened practical understanding of the end-to-end machine learning and deep learning development lifecycle – including data preprocessing, CNN and deep learning model development, model evaluation using industry-standard metrics, and deployment of trained models into usable web applications.
- Hands-on exposure to image processing (OpenCV) and audio signal processing (Librosa, MFCC extraction), and to real-world challenges such as handwriting variability, background noise, and recording-condition variance.
- Successful completion of the CodeAlpha Virtual Internship Program in Machine Learning, along with an official Internship Completion Certificate and Letter of Recommendation.

Aspect	Project 1: Handwritten Digit Recognition	Project 2: Speech Emotion Recognition
Input Type	Image (handwritten digit)	Audio (speech/.wav file)
Core Model	Convolutional Neural Network (CNN)	Deep Learning classification model

Aspect	Project 1: Handwritten Digit Recognition	Project 2: Speech Emotion Recognition
Key Feature Extraction	OpenCV image preprocessing	MFCC extraction using Librosa
Web Framework	Streamlit	Flask
Key Result	99.11% test accuracy on MNIST test set	Predicted emotion with probability breakdown

Table 4.3: Comparison of the two Internship Projects

4.10 Certificates of Completion and Other Communication Mail Proof

On successful completion of the one-month Machine Learning Virtual Internship Program (20th July 2026 to 20th August 2026), CodeAlpha issued an official Internship Completion Certificate (Student ID: CA/DF1/203699) and a Letter of Recommendation, which were sent via official email communication. The certificate and the corresponding email communication proof are attached below.

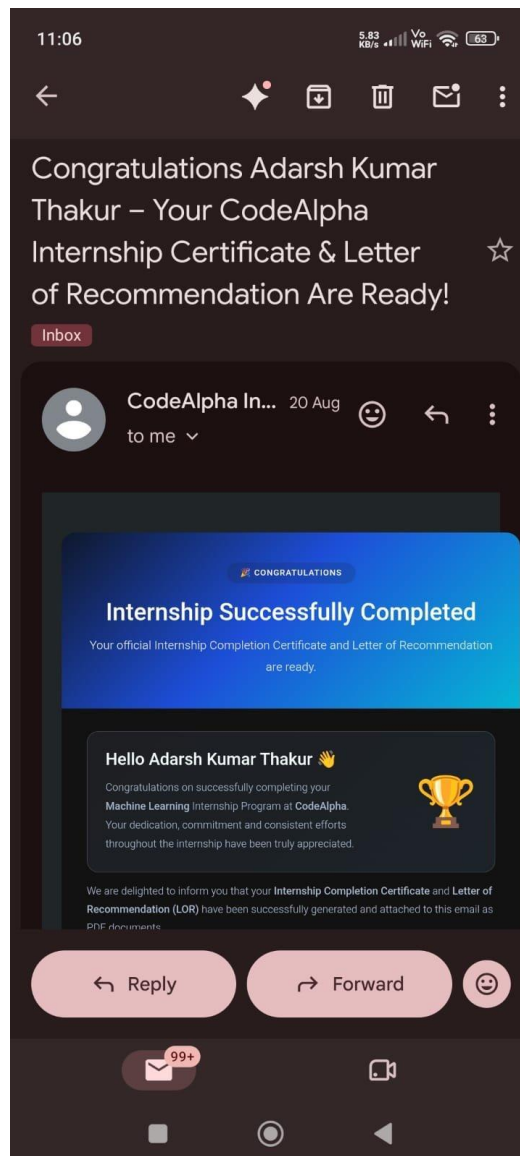


Figure 4.6: Email confirmation received from CodeAlpha Internship team confirming successful completion of the internship



Figure 4.7: Email attachment showing the Internship Completion Certificate and Letter of Recommendation (LOR) issued by CodeAlpha

The complete certificate is also provided separately at the beginning of this report (Figure 3.1, Chapter 3 – Internship Completion Certificate).

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